

Discussion 2: Bad Agents

Wednesday, April 8, 2026

Complete this worksheet in a group of **3-5 students**. Write your answers **by hand**, then take a picture and submit it to Canvas. When you submit, only one person from your group needs to submit, but make sure to specify who is in the group.

Select **one** of the scenarios below, and specify which scenario you are choosing. For **your choice of scenario**, answer the following.

1. Select one scenario (1, 2, or 3).
2. What is your sampling frame? Who are your sampling units?
3. How do you collect your sample, logistically? What type of sampling is it?
4. What information do you collect? How do you measure it?
5. Is your study observational, or experimental? Why?
6. Explain why the way you have chosen to design your study will support your desired conclusion.
7. How would you alter the study to answer the question of interest in a more honest way?

Scenarios (pick **one** of the three)

Scenario 1: Sugar Industry

This is adapted from documented Sugar Research Foundation efforts in the 1960s to emphasize fat and cholesterol as the main dietary causes of coronary heart disease while downplaying sugar. A historical analysis of internal industry documents found that the Foundation sponsored a 1965 literature review, set the review's objective, and steered attention toward fat and away from sucrose. ([Article](#).)

Imagine that you are working for the Sugar Research Foundation in 1965. Some scientists are starting to ask whether eating a lot of sugar increases the risk of heart disease. Your organization is worried that if sugar is blamed, it could hurt sugar sales.

Question of interest: Among adults, which dietary factor is more strongly related to heart disease risk: sugar or fat?

Desired conclusion: Dietary fat and cholesterol are the main diet-related causes of heart disease. Sugar is not an important cause.

Design a study that will likely support your organization's desired conclusion, without fabricating data.

Relevant facts:

- Heart disease tends to take many years to develop.
- Scientists often use blood cholesterol as a risk marker for later heart diseases.
- It is challenging to accurately monitor individuals' diets. There are many ways to go about recording fat and sugar intake.

Scenario 2: Tobacco Industry

The tobacco industry's strategy is very often described not as proving smoking safe, but as manufacturing doubt and uncertainty around the evidence. That general strategy is extensively documented in public health scholarship. ([Article](#).)

Imagine that you are working for a tobacco company in 1962. More scientists are reporting that cigarette smoking may cause lung cancer. Your company knows it may be hard to prove smoking is harmless, so instead it wants a study that makes the evidence look less clear and less convincing.

Question of interest: Among adults, is cigarette smoking clearly associated with respiratory diseases, or could the pattern be explained by other factors?

Desired conclusion: The connection between smoking and respiratory diseases is still uncertain, and other explanations may be just as important.

Design a study that will likely support your industry's desired conclusion, without fabricating data.

Relevant facts:

- Lung cancer and other smoking-related diseases usually develop over many years. Shortness of breath, coughing, etc. tend to show up earlier.
- Factors like air pollution, occupational exposures (asbestos, arsenic, diesel exhaust, etc.), and radon can affect lung cancer risk.

Scenario 3: Health Plan Patient Satisfaction

Patient satisfaction surveys have become widely used by health plans and healthcare organizations as a way to evaluate doctors and compare quality of care. Researchers have also studied how the design of these surveys can affect the results and the conclusions drawn from them. ([Article](#).)

Imagine that you are working for a large health plan. Leadership wants survey results showing that patients are very satisfied with their primary care doctors. These results may be used in reports, advertising, and comparisons across physicians.

Question of interest: Among patients who recently visited a primary care doctor in your health plan, how satisfied are they with the care they received?

Desired conclusion: Most patients are highly satisfied with their care, and satisfaction is high across doctors in the health plan.

Design a study that will likely support your organization's desired conclusion, without fabricating data.

Relevant facts:

- Patient experience surveys like [CAHPS](#) can be done by mail, phone, web, or mixed mode.
- It is common to follow up with individuals who do not respond to these surveys.
- Some organizations use survey methodologists or questionnaire-design experts whose job includes designing and reviewing survey questions.